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Generative AI Engineer — LLM Fine-tuning — RAG Pipelines — Multimodal AI

Professional Summary

Results-driven **Generative AI Engineer** specializing in LLM fine-tuning, Retrieval-Augmented Generation (RAG), and multimodal AI systems. Proven hands-on experience building production-grade GenAI pipelines with LangChain, HuggingFace, and Diffusion Models. Strong foundation in MLOps with FastAPI, Docker, and CI/CD. Passionate about deploying real-world AI solutions that bridge research and production.

Technical Skills

Generative AI: LLM Fine-tuning (QLoRA, LoRA, PEFT), Unsloth, HuggingFace Transformers, LangChain, RAG, Agentic AI, LangGraph, Prompt Engineering, Stable Diffusion, Diffusers, ControlNet, VAE, DCGAN
LLMs & Multimodal: LLaMA 3, Mistral, Gemini API, OpenAI API, LLaVA, Qwen-VL, BERT, RoBERTa, Watsonx
Vector Stores & Search: ChromaDB, FAISS, Qdrant, Pinecone, Semantic Search, Reranking
ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, SHAP, SMOTE, EfficientNet, YOLOv8
MLOps & Backend: FastAPI, Flask, Docker, GitHub Actions, CI/CD, MLflow, Streamlit, Gradio, Kubernetes
Cloud & Data: Microsoft Azure, AWS (Basic), PostgreSQL, MySQL, SQLite, Pandas, NumPy
Tools: Git, GitHub, VS Code, Jupyter, Linux CLI, Bash Scripting, SSH

Work Experience

Software Engineer Intern / Backend Developer Intern

Algotix AI

Jan 2026 – Present

Bahawalpur, Pakistan

- Developing and maintaining backend services and REST APIs using Python and FastAPI to support AI-driven product features.
- Collaborating with cross-functional teams to design scalable system architectures for production-grade deployments.
- Contributing to intelligent automation workflow infrastructure with robust error handling and logging systems.

Machine Learning Engineer

Deep Embed Lab

Nov 2024 – Jan 2026

Bahawalpur, Pakistan

- Developed and optimized predictive ML models using Python, Scikit-learn, and TensorFlow, improving prediction accuracy by **15%**.
- Automated daily data ingestion and processing pipelines using Linux Shell Scripting and Cron jobs, reducing manual effort by **30%**.
- Performed rigorous EDA on large datasets and visualized trends with Matplotlib/Seaborn for non-technical stakeholders.

Data Science Intern

Arch Technologies

Nov 2025 – Dec 2025

Islamabad, Pakistan

- Designed intelligent systems using Python frameworks to build autonomous agents capable of multi-step planning and decision-making.
- Developed production-grade RAG pipelines with FastAPI and vector databases, achieving high retrieval accuracy and low latency.
- Containerized services with Docker and deployed scalable prototypes on Linux-based cloud environments.

Data Scientist

CodeAlpha

Oct 2025 – Nov 2025

Remote, India

- Conducted data cleaning, preprocessing, and EDA using Python to significantly enhance data quality for downstream ML tasks.
- Developed and evaluated ML models for classification and prediction, delivering measurable improvements in analytical capabilities.

Key Projects

- MRI Brain Tumor Detection & Classification** | *EfficientNetB2 · Tensorflow · Grad-CAM · Gradio · HuggingFace*
- Built a 4-class tumor classification system (Glioma, Meningioma, Pituitary, No Tumor) using **EfficientNetB2** transfer learning on 7,023 Kaggle MRI images, achieving **97%+ validation accuracy**.
 - Integrated **Grad-CAM** explainability overlays for clinical interpretability; deployed as a **Gradio** app on HuggingFace Spaces with real-time inference on uploaded scans.
- Multimodal LLM Fine-tuning — Visual Document Understanding** | *Qwen2-VL · LoRA · PEFT · HuggingFace*
- Fine-tuned **Qwen2-VL (7B)**, a vision-language model, on a custom invoice and medical report dataset using **LoRA** adapters via PEFT, enabling structured extraction of text and tables directly from document images.
 - Achieved significant improvement in field-level extraction accuracy over zero-shot baseline by training on domain-specific image-text instruction pairs with mixed-modal inputs.
 - Deployed the fine-tuned multimodal model to **HuggingFace Spaces** with a **Gradio** interface supporting image upload and natural language querying over document contents.
- QLoRA Fine-tuned LLM for Domain Q&A** | *LLaMA 3 · Unsloth · PEFT · HuggingFace · Kaggle GPU*
- Fine-tuned **LLaMA 3 (8B)** on a domain-specific instruction dataset using **QLoRA** via Unsloth, cutting VRAM usage by **70%** to run entirely on free Kaggle T4 GPUs with 4-bit quantization.
 - Merged PEFT adapters into base model and published to **HuggingFace Hub** with an interactive Spaces demo for real-time inference.
- Multi-PDF RAG Research Assistant** | *LangChain · ChromaDB · Gemini API · Streamlit*
- Built an end-to-end **RAG pipeline** ingesting multiple PDFs, embedding chunks into **ChromaDB** vector store, and answering natural language queries via Gemini API with source citation.
 - Implemented semantic search with cosine similarity reranking; deployed as a **Streamlit** web app with session-based conversation history and multi-document Q&A.
- Stable Diffusion Fine-tuning & Image Synthesis App** | *Diffusers · DreamBooth · ControlNet · VAE · Gradio*
- Fine-tuned **Stable Diffusion v1.5** on a domain-specific image dataset using **DreamBooth** to generate high-fidelity synthetic images; integrated **ControlNet** for pose and edge-map conditioned generation.
 - Built a custom **VAE**-based anomaly detector using reconstruction error thresholding on latent embeddings, applicable to defect detection in manufacturing and medical imaging.
 - Wrapped the full pipeline — text-to-image, image-to-image, and inpainting — in a **Gradio** app deployed on HuggingFace Spaces with prompt control and one-click image export.
- End-to-End MLOps Pipeline** | *GitHub Actions · Docker · FastAPI · MLflow · CI/CD · GHCR*
- Built a complete **CI/CD pipeline** using GitHub Actions to auto-build and push Docker images to GitHub Container Registry on every commit, with automated unit testing and model artifact versioning via **MLflow**.
 - Deployed a **FastAPI** REST endpoint for real-time ML inference with preprocessing pipelines, health checks, and structured logging.

Education

Islamia University of Bahawalpur
Bachelor of Science in Computer Science

Jun 2022 – Jun 2026
Bahawalpur, Pakistan

Certifications

- **Oracle Generative AI Professional** (1Z0-1127-25) — Oracle University
- **IBM Generative AI Engineering** with Python, LangChain & Watsonx — Coursera (IBM Verified)
- **IBM AI Engineering** with Python, PyTorch & TensorFlow — Coursera (IBM Verified)
- **Machine Learning Specialization** — Coursera (Stanford / DeepLearning.AI)
- **Natural Language Processing Specialization** — Coursera (DeepLearning.AI)
- **AI Engineer for Data Scientists** (Associate) — DataCamp
- **Microsoft AI and ML Engineering** — Microsoft
- **Gemini Certified University Student** — Google
- **Introduction to Kubernetes** (LFS158) — Linux Foundation